

TimeLightHub — Phase-1 Protocol: the single-cell experiment

Public summary for partner laboratories · Samer Beyrouti, Inventor & Architect · Connect@TimeLightHub.com · ORCID 0009-0003-8565-2605 · Zenodo DOI 10.5281/zenodo.22649324 · Patent Pending — GB2615417.9 (filed 18 June 2026) · September 2026

1. Purpose

TimeLightHub is a memory-centric photonic computing architecture whose energy claim reduces to one measurable quantity: **the energy to read one bit stored in a light-switchable (photochromic) memory cell**. Phase 1 builds a single such cell and measures that quantity, together with the cell's readout latency. The result is publishable on its own, whatever it shows, and it decides whether the rest of the architecture is worth building.

The experiment in one line. Fabricate one diarylethene (DAE) memory cell in a transparent polymer host, **write at 375 nm** (open → closed isomer), **read at 405 nm** (non-destructive probe of the stored state), **reset at 532 nm** (closed → open), and measure **(i) readout latency** and **(ii) switching energy per bit** for write, read and reset.

2. The cell

A thin film of a diarylethene photochromic compound dispersed in a polymer host (SU-8 or PMMA class; the host and DAE derivative to be chosen with the partner from what the lab already handles). Stage A characterises the film on a planar substrate; Stage B (optional in Phase 1) repeats the readout inside a simple channel waveguide so the cell is probed the way the architecture will use it. No custom photonic process is required for Stage A; Stage B fits an existing polymer-waveguide or SiN platform.

3. Measurements

Quantity	How it is obtained	Why it matters
M1 · Readout latency	Pulsed 405 nm probe; time-resolved detection of the transmitted/reflected signal versus stored state. Report the shortest probe pulse that still resolves the two states at a stated contrast.	Sets the clock of the memory bank and of the AI engine's weight reads.
M2 · Switching energy per bit	Write and reset: minimum optical dose (pulse energy × spot area → energy per cell) that drives a full, repeatable state change. Read: probe energy per resolvable read at the contrast used in M1.	The deciding number. Public energy window: tenths of a pJ per read = large advantage; a few pJ narrows it; above ten it disappears.
Supporting	Optical contrast between states at 405 nm; cycle endurance (write/reset cycles before contrast falls by a stated fraction); dark retention of the written state over days.	Confirms the cell is a usable non-volatile latch, not only a switch.

4. Method outline

- **Prepare** the DAE-doped film (spin-coat; thickness and loading chosen with the partner).
- **Characterise** absorption spectra of open and closed states; confirm 375 / 405 / 532 nm selectivity.
- **Write** with 375 nm pulses of increasing dose; record the dose–response curve to saturation.
- **Read** with 405 nm pulses of decreasing energy and duration until the states no longer separate → M1 and read-energy part of M2.
- **Reset** with 532 nm; repeat the cycle to obtain endurance; hold written cells in the dark for retention.
- **Report** M1, M2 and the supporting metrics with uncertainties; compare with the simulator's predicted window.

5. Success gates, scope and roles

Gate	Outcome
Go	Repeatable write/read/reset with clear contrast; read energy within the public window ($\leq$ a few pJ per bit) — proceed to Stage 2 (one byte and the first optical lookup).
Learn	Cell works but read energy is above the window — the result is still publishable; the architecture's energy claim is revised with real numbers.
Stop	No repeatable non-volatile switching in the chosen host — reported openly.

Scope: months, not years; materials-scale, not program-scale; an ideal graduate project, scoped together with the partner. **Partner provides:** materials handling, film preparation, the three sources (375 / 405 / 532 nm), time-resolved detection, and the student or staff time. **TimeLightHub provides:** the architecture and the public papers; under a simple mutual NDA, the element-physics dossier, the node specification, and the physics-honest simulator with its predicted targets for M1 and M2; co-authorship on the resulting paper is expected and welcome.